

up the window and keeps only the title bar visible. Or, he can add the drag-and-drop feature to an application.

The X Window System is a system-level application, or rather, a process that handles the display in Linux. Basically, what you see on your screen is coming from X. X “talks” to the graphics display device (the graphics processor) and tells it what to display.

As X interacts with the hardware at a basic level, programmers need not be bothered about giving instructions to the video card for any display (mind you, the GUI taxes a system heavily). With the difficult job being taken care of, programmers only need to give instructions to X saying “move window 1 to position x” or “minimise window 2” or “refresh window 3 every 5 seconds”.

The X concept came about when a GUI for UNIX was being worked out. It was released in 1984 by the *Athena Project*, an academic project undertaken at the Massachusetts Institute of Technology. It was taken over in 1998 by the *X Consortium*, which has been maintaining the standards for X ever since. The X specifications are freely available for further development, and Linux developers adopted these specifications and called it the Xfree86.

Xfree86, like its base—the X—is very versatile, with various options that developers can use to make different windows (or apps) display differently. X follows a

client-server architecture.

X is the server, and the apps that need display access are clients. This is a client-server model because X can control windows on a remote PC via a LAN

where the user is on a *dumb* (CPU-less) system. X can thus cater to several users simultaneously. Even on a standalone system, X acts as a server, and the desktop environments and applications act as clients.

